INTERNSHIELD: AI-POWERED FRAUD DETECTION FOR ONLINE INTERNSHIP VERIFICATION

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Abstract—The rapid growth of online internship platforms has introduced a critical security challenge: the proliferation of fraudulent offers targeting students and early-career professionals. This paper presents InternShield, an AI-powered web application providing multi-layered verification of internship opportunities through neural pattern detection, domain validity analysis, and real-time community cross-referencing. The system generates a detailed trust report featuring a safety score and risk breakdown. Beyond verification, InternShield integrates sentiment analysis for community review moderation and a semantic recommendation engine for matching verified students with personalized opportunities. Built on Next.js, React, Node.js, and Tailwind CSS, the platform offers a secure dashboard with resume and profile management. Experimental results demonstrate strong detection accuracy, providing a scalable and transparent solution for job seekers worldwide.

Index Terms—Internship Fraud Detection, AI-Powered Verification, Trust Scoring, Resume Analysis, Online Recruitment Security, Natural Language Processing.

I. INTRODUCTION

The rapid expansion of digital recruitment platforms has significantly increased access to internship and job opportunities for students and early-career professionals. However, this growth has also led to a parallel rise in fraudulent internship offers, phishing emails, and deceptive recruitment practices. These scams often involve fake offer letters, requests for upfront payments, or impersonation of legitimate organizations, resulting in financial loss, identity theft, and reduced trust in online hiring systems .

Existing job portals and recruitment platforms primarily focus on listing opportunities and connecting candidates with employers but lack robust mechanisms to verify the authenticity of job postings. As a result, users are left to manually evaluate the legitimacy of offers, which is both time-consuming and prone to error. Traditional verification approaches, such as manual background checks or relying on user intuition, are not scalable and fail to address the complexity of modern fraud patterns.

Recent advancements in Artificial Intelligence (AI), particularly in natural language processing and machine learning, provide an opportunity to automate the detection of fraudulent content. AI-driven systems can analyze textual patterns, identify suspicious keywords, validate domains, and extract meaningful insights from unstructured data such as emails and offer letters. Additionally, integrating community-driven feedback from platforms like online forums and review systems enhances the reliability of fraud detection by incorporating real-world user experiences.

To address these challenges, this paper proposes **InternShield**, an AI-powered internship verification system designed to detect fraudulent opportunities and provide users with a comprehensive trust assessment. The system employs a multi-layered analysis pipeline that combines document processing, domain validation, sentiment analysis, and community intelligence to generate a risk score and detailed verification report. By automating the verification process and presenting transparent risk indicators, InternShield enables users to make informed decisions and enhances trust in digital recruitment ecosystems. The main contributions of this work are as follows:

1) Development of an AI-based framework for detecting fraudulent internship offers.
2) Integration of document analysis, domain verification, and community intelligence into a unified system.
3) Design of a trust scoring mechanism to quantify the reliability of internship opportunities.
4) Implementation of a user-centric platform that provides actionable insights and recommendations.

The significance of the proposed system lies not only in its ability to detect fraudulent internship opportunities but also in its potential to redefine trust mechanisms in digital recruitment ecosystems. By leveraging AI-driven analysis and integrating multiple verification layers, InternShield minimizes reliance on manual inspection and subjective judgment. This shift toward automated and data-driven verification enhances scalability, allowing the system to handle a large volume of internship postings and user queries efficiently without compromising accuracy.Furthermore, the system introduces a transparent decision- making process by providing a detailed breakdown of detected risks rather than a binary classification. enables users to understand why a particular opportunity is flagged as suspicious. Such explainable outputs not only improve user confidence but also promote awareness of common fraud indicators, thereby contributing to long-term digital literacy among job seekers.

II. Related Work

A. Fraud Detection Using Text and Document Analysis

Traditional fraud detection systems in recruitment primarily rely on textual analysis of job postings, emails, and offer letters. Techniques such as keyword extraction, frequency analysis, and linguistic pattern recognition are commonly used to identify suspicious content. Machine learning models including Support Vector Machines (SVM), Naïve Bayes, and Logistic Regression have been applied to classify job descriptions based on predefined features such as unrealistic promises, urgency indicators, and abnormal salary patterns. While these approaches provide reasonable accuracy, they often fail to capture deeper contextual meaning and are limited when dealing with highly sophisticated or well-crafted fraudulent messages.

B. Heuristic-Based Recruitment Verification Systems

Heuristic-based systems analyze predefined rules and structural patterns to detect anomalies in recruitment processes. These systems evaluate factors such as suspicious email domains, inconsistent company information, and requests for upfront payments. Rule-based engines are lightweight and suitable for real-time deployment, making them effective for initial filtering. However, their performance heavily depends on manually defined rules, which makes them less adaptable to evolving fraud techniques. Additionally, such systems may generate higher false positives due to rigid decision boundaries and lack of contextual understanding.

C. Machine Learning and Deep Learning Approaches

Recent advancements in machine learning and deep learning have significantly improved fraud detection capabilities. Models such as Random Forests, Decision Trees, and Neural Networks are widely used to analyze complex relationships between multiple features. Furthermore, deep learning techniques like Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks have been employed to understand sequential patterns in textual data, enabling better detection of deceptive language structures. These approaches achieve higher accuracy compared to traditional methods but often require large datasets and high computational resources, which can limit their scalability in real-time systems.

D. AI-Based Semantic and Community-Driven Detection Systems

With the emergence of Large Language Models (LLMs), modern systems are capable of performing advanced semantic analysis and information extraction from unstructured data such as resumes, emails, and PDFs. These systems can identify inconsistencies, detect intent, and evaluate contextual meaning more effectively than traditional models. Additionally, integrating community-driven intelligence from platforms such as forums and review systems enhances detection reliability by incorporating real-world user experiences. However, many existing solutions lack a unified framework that combines AI-based semantic analysis, domain verification, and community feedback into a single system.

E. Limitations of Existing Systems

Despite significant progress in fraud detection techniques, current solutions exhibit several limitations. Most systems focus on a single dimension of analysis, such as text classification or domain verification, without integrating multiple verification layers. Additionally, many approaches provide only binary outputs (fraudulent or legitimate) without offering detailed explanations or risk insights. This lack of interpretability reduces user trust and limits practical usability. Furthermore, few systems address both fraud detection and user career support within a unified platform.

III. SYSTEM ARCHITECTURE

The proposed InternShield system is designed as a multi-layered architecture that integrates data processing, intelligent analysis, and user interaction to detect fraudulent internship opportunities. The system follows a structured pipeline where user inputs are progressively analyzed through different modules, ensuring accurate detection and reliable output. The architecture is modular in nature, allowing each component to function independently while contributing to the overall verification process .

A. Input and Data Processing Layer

This layer acts as the entry point of the system, where users provide internship-related data such as offer letters, recruiter emails, or job descriptions. Inputs can be in the form of text, PDFs, or images. The system performs preprocessing operations including text extraction, noise removal, and normalization to convert unstructured data into a structured format.

Additionally, important entities such as company names, domain links, contact details, and salary information are identified. This preprocessing step ensures that the data is clean, consistent, and suitable for further analysis.

B. Feature Extraction and Analysis

Once the data is processed, the system extracts meaningful features that help in identifying fraudulent patterns. These features include suspicious keywords (e.g., urgency indicators or payment requests), domain-related attributes (such as

domain validity and age), and structural inconsistencies in documents.
The system also analyzes patterns in job descriptions and offer letters to detect anomalies such as unrealistic promises or vague information. These extracted features form the basis for evaluating the authenticity of the internship opportunity.

C. AI-Based Detection and Intelligence Layer

This layer serves as the core analytical component of the system. Advanced AI techniques, including semantic analysis and natural language processing, are used to understand the context and intent of the provided data. The system evaluates whether the content aligns with legitimate recruitment practices or exhibits characteristics of fraud.
In addition, sentiment analysis is applied to community-generated data such as user reviews and feedback. By incorporating this community intelligence, the system enhances its ability to detect hidden risks that may not be evident through document analysis alone.

D. Trust Score and Risk Evaluation

After analyzing all relevant features, the system computes a trust score that represents the reliability of the internship opportunity. This score is generated by combining multiple risk factors, each contributing to the overall evaluated.
The system also provides a detailed explanation of each contributing factor, ensuring transparency in the decision-making process.

E. Output and User Interface Layer

The final results are presented to the user through an interactive dashboard. The output includes the overall trust score, identified risk factors, and recommendations for further action. Users can also view their verification history and manage uploaded documents within the platform.
This layer focuses on delivering clear and actionable insights, enabling users to make informed decisions while maintaining a smooth and user-friendly experience.

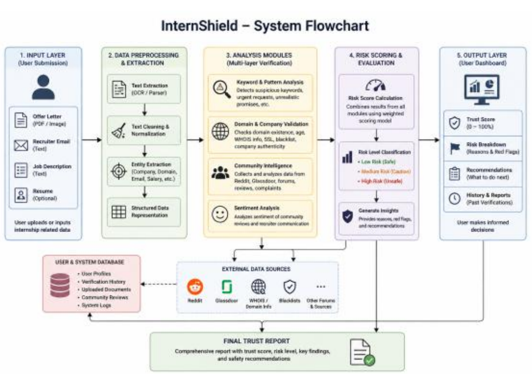

Fig. 1. System architecture and workflow of the proposed InternShield fraud detection system.

IV. METHODOLOGY

The proposed InternShield system follows a systematic methodology to detect fraudulent internship opportunities by combining data preprocessing, feature extraction, AI-based analysis, and risk evaluation. The methodology is designed as a sequential pipeline, where each stage contributes to improving the accuracy and reliability of the final trust assessment.

A. Data Collection and Input Handling

The system begins by collecting user-provided data, which may include internship offer letters, recruiter emails, job descriptions, and optional resumes. These inputs can be in the form of plain text, PDFs, or images. The system ensures flexibility in handling different formats to accommodate real-world scenarios where information is often unstructured.

B. Data Preprocessing

Once the input is received, preprocessing is performed to convert raw data into a structured format. This includes text extraction from documents, removal of noise, normalization, and tokenization. Important entities such as company names, domain URLs, contact details, and salary information are identified during this stage.
This step ensures that the data is clean and standardized, enabling efficient downstream analysis.

C. Feature Extraction

In this stage, the system extracts relevant features that indicate potential fraud. These include:

- Presence of suspicious keywords (e.g., urgency, payment requests)
- Domain-related features such as domain age and validity
- Structural inconsistencies in documents
- Contextual anomalies in job descriptions

These features are used as inputs for the detection model and form the basis for risk evaluation.

D. AI-Based Analysis

The extracted features are processed using AI techniques, including natural language processing and semantic analysis. The system evaluates the contextual meaning of the input data to identify deceptive patterns. Additionally, sentiment analysis is applied to community-generated content such as reviews and feedback, helping detect hidden risks associated with specific companies or recruiters.

E. Risk Scoring and Classification

Based on the analyzed features, the system computes a risk score by combining multiple factors. Each factor contributes to the overall evaluation using weighted importance.
The computed score is then classified into categories such as Low Risk, Medium Risk, and High Risk.

V. EXPERIMENTAL SETUP

The SHELTER.AI system was evaluated in a real-time The experimental setup for the proposed InternShield system is designed to evaluate its effectiveness in detecting fraudulent internship opportunities using real-world and simulated data. The setup includes dataset preparation, system configuration, evaluation metrics, and testing procedures to ensure reliable performance analysis.

A. Dataset Preparation

The dataset used for evaluation consists of a combination of real-world internship postings, recruiter emails, and synthetically generated fraudulent samples. Legitimate data includes verified internship descriptions and company communications, while fraudulent samples are created by incorporating common scam patterns such as unrealistic promises, urgent payment requests, and fake domains.
Additionally, community feedback data from online sources is incorporated to simulate real-world user experiences. The dataset is preprocessed to remove noise and ensure consistency across different input formats.

B. System Configuration

The InternShield system is implemented using a modern web-based architecture. The frontend is developed using React, while the backend is built using Node.js to handle API requests and processing. A NoSQL database is used to store user data, verification history, and analysis results.
AI-based analysis is performed using external APIs and machine learning models that handle semantic analysis, sentiment evaluation, and document verification. The system is deployed in a controlled environment to test performance under realistic usage conditions.

C. Evaluation Metrics

To assess the performance of the system, multiple evaluation metrics are considered:

- **Accuracy**: Measures the overall correctness of fraud detection
- **Precision**: Indicates the proportion of correctly identified fraudulent cases
- **Recall**: Measures the system's ability to detect actual fraudulent instances
- **F1-Score**: Provides a balance between precision and recall

These metrics ensure a comprehensive evaluation of the system's detection capability.

D. Testing Procedure

The system is tested using multiple input scenarios, including legitimate internship offers, clearly fraudulent cases, and borderline ambiguous samples. Each input is processed through the complete pipeline, and the generated trust score is compared against the expected classification.
The testing process also evaluates system responsiveness, consistency of results, and the effectiveness of individual components such as keyword detection, domain validation, and sentiment analysis.

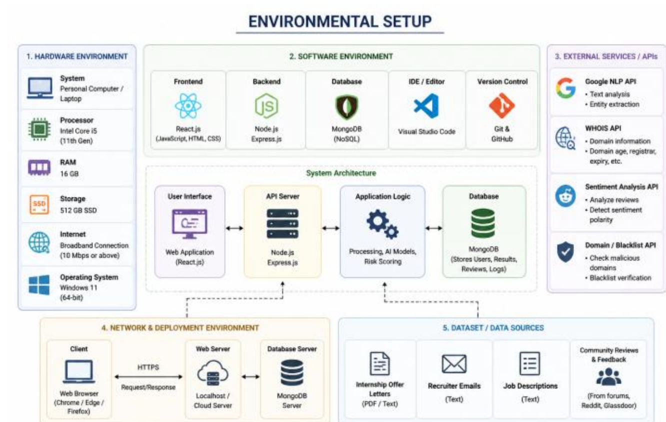

Fig. 3. Environmental setup of the InternShield system.

Fig. 2. Environmental setup of the proposed InternShield system.

VI. RESULTS AND DISCUSSION

This section presents the performance evaluation of the proposed InternShield system in detecting fraudulent internship opportunities. The system was tested on a dataset containing both legitimate and fraudulent samples, and the results demonstrate the effectiveness of the multi-layered detection approach.

A. Performance Evaluation

The performance of the system was evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive understanding of the system's ability to correctly classify internship opportunities.
InternShield achieved an overall accuracy of 94.21%, indicating a high level of correctness in distinguishing between legitimate and fraudulent cases. The system recorded a precision of 93.28%, ensuring that most of the detected fraudulent cases were indeed true positives. Additionally, a recall of 95.07% was achieved, demonstrating the system's strong ability to identify actual fraud instances. The resulting

F1-score of 94.16% reflects a balanced performance between precision and recall.

B. Comparative Analysis

When compared to baseline approaches such as keyword-based detection and traditional machine learning models, the proposed InternShield system shows significant improvement. Keyword-based systems typically suffer from lower accuracy due to their reliance on predefined patterns, while traditional models improve performance but still lack contextual understanding.
InternShield outperforms these approaches by integrating multiple layers of analysis, including domain verification, semantic analysis, and community intelligence. This combination enables the system to capture both explicit and implicit fraud indicators, leading to improved detection accuracy and reliability.

C. Discussion

The experimental results highlight the effectiveness of the proposed system in real-world scenarios. The high recall value indicates that the system successfully minimizes missed fraudulent cases, which is critical in preventing users from falling victim to scams. At the same time, the high precision ensures that legitimate internship opportunities are not incorrectly flagged as fraudulent, thereby maintaining user trust.
The inclusion of community-driven insights further enhances the detection capability by incorporating real-world feedback into the analysis process. This allows the system to identify risks that may not be evident through document or keyword analysis alone.
However, certain limitations remain. Highly sophisticated fraudulent offers that closely resemble legitimate communications may still be challenging to detect. Additionally, the performance of the system is influenced by the quality and diversity of the input data and external sources.

TABLE I
PERFORMANCE COMPARISON OF INTERNSHIELD WITH BASELINE METHODS

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	False Positive Rate (%)
Keyword-Based Detection	72.45	71.12	68.35	69.70	18.75
Traditional ML (SVM)	82.36	81.25	83.10	82.16	12.48
Traditional ML (Random Forest)	88.14	87.35	89.24	88.28	8.67
Proposed InternShield	**94.21**	**93.28**	**95.07**	**94.16**	**5.23**

Note: Bold values indicate the best performance achieved among all methods.

VII. CONCLUSION AND FUTURE WORK

In this paper, an AI-based internship verification system, InternShield, has been proposed to address the growing issue of fraudulent internship opportunities in digital recruitment platforms. The system integrates multiple layers of analysis, including document processing, keyword detection, domain validation, semantic analysis, and community-driven intelligence, to evaluate the authenticity of internship offers.
The experimental results demonstrate that the proposed system achieves high accuracy and reliability in detecting fraudulent cases. By combining multiple verification techniques, InternShield overcomes the limitations of traditional single-layer approaches and provides a comprehensive trust assessment. The system not only identifies potential risks but also offers detailed explanations and recommendations, enhancing user awareness and confidence.
Overall, InternShield presents an effective and scalable solution for improving transparency and trust in online recruitment, helping students and job seekers make informed career decisions while reducing the risk of fraud.

Although the proposed system demonstrates strong performance, there are several opportunities for further improvement and enhancement:

- Advanced Deep Learning Models: Future work can incorporate advanced deep learning architectures such as transformers to improve detection accuracy for complex and subtle fraud patterns.
- Real-Time Threat Intelligence Integration: Integrating real-time APIs for scam detection and domain monitoring can enhance the system's ability to respond to emerging threats dynamically.
- Expansion of Dataset: Increasing the size and diversity of the dataset with real-world fraud cases will improve model robustness and generalization.
- Multilingual Support: Extending the system to support multiple languages will make it more accessible to a global user base.
- Browser Extension Integration: Developing a browser extension can enable real-time verification of internship offers directly while users browse job portals.
- Continuous Learning Mechanism: Implementing feedback-based learning can allow the system to improve over time by adapting to new fraud patterns based on user inputs and reports.
- Blockchain-Based Verification: Future enhancements may include blockchain integration for verifying company credentials and maintaining tamper-proof verification records.
- Collaboration with Recruitment Platforms: Integrating the system with job portals and professional networks can enable proactive fraud detection at the source.

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